

Area as Tiling Mission

Warm-up and worked example

Name: _____

Date: _____

PAGE 1 | WEEK 8, DAY 2

YOUR MISSION Understand area as the number of square units needed to cover a flat region without gaps or overlaps.

SMART MOVE Name the attribute being measured before choosing a method.

Math Talk

Which covers more space: a long skinny rectangle or a short wide rectangle? What information do you need?

My idea:

New Words

attribute	parallel	perpendicular
My meaning or picture:	My meaning or picture:	My meaning or picture:

Worked Example

Do this example with your teacher. Say what changes and what stays the same.

1. Cover rectangles with square tiles.
2. Count tiles by rows instead of one by one.

What the example shows: _____

Area as Tiling Mission

Learn it and show your thinking

Name: _____

Date: _____

PAGE 2 | WEEK 8, DAY 2

YOUR MISSION Follow the learning path, then explain the big idea.

SMART MOVE Draw before asking for a rule.

[] **Step 1:** Cover rectangles with square tiles.

[] **Step 2:** Count tiles by rows instead of one by one.

[] **Step 3:** Discuss why units must be equal squares.

[] **Step 4:** Tile irregular rectilinear shapes.

[] **Step 5:** Connect area to "covering" rather than "around."

My model, drawing, or organized work:

The big idea in my own words:

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Practice lab

Name: _____

Date: _____

PAGE 3 | WEEK 8, DAY 2

YOUR MISSION Use a model, equation, or explanation for every task.

SMART MOVE Check whether your answer is reasonable.

1. Count area by tiling.

2. Draw shapes with area 12 square units.

3. Compare two shapes by area.

4. Fix a tiling with gaps or overlaps.

Choose one answer and prove it another way:

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Challenge and final check

Name: _____

Date: _____

PAGE 4 | WEEK 8, DAY 2

YOUR MISSION Try an unfamiliar problem and defend your reasoning.

SMART MOVE A wrong first idea can still lead to a strong solution.

Challenge Mission

Draw three different shapes with area 16 square units.

My plan:

My solution and proof:

Final Check

Why is area measured in square units?

☐ I can teach it

☐ I need practice

☐ I need help

Area as Tiling Mission

Reusable math tool

Name: _____

Date: _____

PAGE 5 | WEEK 8, DAY 2

YOUR MISSION Use this page to draw, model, organize, and check.

SMART MOVE Name the attribute being measured before choosing a method.

Grid Lab

What does your drawing prove?
